

Unit Testing with Junit

1. Basic Test Case

Question:

Write a JUnit 5 test case to verify that the sum of two integers using a `Calculator.add(int a, int b)` method returns the correct result.

2. Test Exception Handling

Question:

Write a test case using `assertThrows` to verify that a method `Calculator.divide(int a, int b)` throws `ArithmeticException` when dividing by zero.

3. Setup and Teardown

Question:

Write a test class that uses `@BeforeEach` and `@AfterEach` to set up and tear down test data for each test method. Print messages during setup and teardown to confirm the execution order.

4. Test Object Equality

Question:

Write a test using `assertEquals` to compare two `Person` objects with the same state. Override `equals()` and `hashCode()` if needed.

5. Test Multiple Assertions

Question:

Write a single test method that uses `assertAll()` to validate multiple conditions: a string is not null, not empty, and contains a specific word.

6. Disabled Test

Question:

Write a test method annotated with `@Disabled` and add a reason. Demonstrate that the test is skipped when running the test suite.

